Project Proposal

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- Name: Pachinkox
- Topic: Simulation/Game
- Repo: https://github.com/tannerweber/pachinkox.git

This project will perform a basic simulation of a pachinko machine. A ball will be dropped, hit pins, then have an ending position on the ground.

Theme

A pachinko machine involves a ball dropping then bouncing off a variety of pins that change its path. The simple behavior of moving left or right when hitting a pin leads to unpredictability that transforms a small difference in the starting position to different ending positions, which makes it interesting.

Rules

I will implement very basic 2D physics that deal with collisions of the ball on fixed points and flat walls on each side of the space, as well as gravity, which can just be a simple decrement to y velocity of the ball each frame.

Scope

If I have issues with this, I could also create a non-physics based algorithm that takes positions of pins in a row and ball information as inputs then returns a new position for the ball to start the next row at until all rows have been passed.

Crates, like `rapier2d`, exist for physics, but I think this may make the project too small.

Architecture

Input parameters to the simulation can be the ball starting position and pin positions. The main output will be the final position and pins touched.

A library crate can perform the simulation and a binary crate can be used to help visualize the results. Basic text output could show each pin hit by the ball to visualize the path.